

PSP Oracle Exit - Monolith Database Migration Tiptoe Runbook (Oracle Hash)

- [Goal](#)
- [Tiptoe Summary](#)
 - [Status](#)
 - [Tiptoe Progress](#)
 - [Duration](#)
 - [Tiptoe Highlights](#)
 - [Business Metrics](#)
 - [Functional Metrics](#)
 - [Performance Metrics](#)
 - [Database Metrics](#)
 - [Snoozing A2i](#)
 - [Downtime Details](#)
 - [Downtime Summary](#)
 - [Learnings/Enhancements](#)
- [Important Resources](#)
 - [Dev Portal](#)
 - [JSS](#)
 - [Splunk](#)
 - [ArgoCd](#)
 - [Notification Email Addresses](#)
- [Change Requests](#)
- [Runbook](#)
 - [Pre Tiptoe Steps](#)
 - [TipToe Steps](#)
 - [Post TipToe Steps](#)
 - [Rollback Decision and Action Plan](#)
 - [Rollback Steps](#)
 - [Post Rollback Steps](#)
- [Monitoring](#)
 - [Apps Health](#)
 - [PSP](#)
 - [PAPI](#)
 - [Pubsub](#)
 - [Report Job Framework](#)
 - [PayrollDM Monitoring](#)
 - [Performance](#)
 - [Hourly Monitoring](#)
 - [Tiptoe Batch Jobs Monitoring](#)
 - [BatchJobs in Activity Window](#)
 - [Critical Batch Jobs Monitoring](#)
 - [Oracle Hash Specific Monitoring](#)
 - [Workflows Monitoring](#)
 - [DB Queries Monitoring](#)
 - [Database Monitoring](#)
 - [Infrastructure Monitoring](#)
 - [Golden Gate Signals](#)
 - [Wavefront](#)
 - [AppDynamics](#)
- [Failed/Missed Batch Jobs](#)

Goal

- Validate Deployment, App and Database Re-architecture changes for Oracle Hash Migration.
- Understand and learn the system behaviour functional/non functional in the Oracle Hash database.
- Validate Application Performance and Database Performance/Stability.
- Validate 50% of P0 OLTP workflows and 35% of OLAP workflows.
- Validate the system under less load i.e. Weekend Traffic will have minimal impact.
- Validate Planned Rollback to prepare for any emergency Rollback.
- Validate all the above goals with the minimal customer and partner team impacts.

Tiptoe Summary

Status

Environment	Status
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Production	COMPLETED
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Tiptoe Progress

JIRA	PSP-27731 - Getting issue details... STATUS
Slack Channel	#psp-monolith-oracle-hash-prod-tiptoe

Duration

Tiptoe Start Time	Tiptoe End Time	Duration
27 Aug 2023 10AM IST	28 Aug 2023 1PM IST	~24 Hours

Tiptoe Highlights

Business Metrics

- **19K payroll runs** completed and no failures observed.
- **18K employees** viewed paychecks via Workforce Portal.
- **700K+ OLTP requests** served across services.
- **3.5K batch job runs** completed and **21 unique batch jobs** completed successfully.
- **71K+ paycheck/bank-account events** were published for risk assessment.
- **42 Reports** executed and **34 unique reports** completed.
- **3 Access Checks** and **1 Delete Requests** were served by Desktop Payroll DM (Data Governance).
- **Performance** metrics was on par with the Oracle Range instance.
- **Replication Lag is < 10s** between Oracle Range and Oracle Hash instances, and **data parity** is green.
- Application and Database infrastructure, stability and performance metrics look good.

Functional Metrics

- **60% of OLTP workflows** are validated. This excludes care Agent Portal workflows.
- **25% of OLAP Batch workloads** are validated.
- **60% of Biz Ops Reports** are validated.

Performance Metrics

- OLTP Workflows performed on par and the performance deviation was ~10%, within the SLA.
- OLAP Workflows performed on par and deviation was < 30%, within the SLA.
- Performances will improve over time once the database warms up.

Database Metrics

- Average Active sessions – 30, Max DB connections - 1385
- Average CPU utilisation - 2.5%
- IOPS
 - Average **Read IOPS** usage was ~9k, max at 18.5k
 - Average **Write IOPS** usage was 422, max at 926
- IO Latency:
 - **Read Latency** < 2ms
 - **Write Latency** < 4ms

Snoozing A2i

S. No	Application	A2I Maintenance URL
1	DT-Payroll-Svc	https://devportal.intuit.com/app/dp/resource/6847740825712235356/incidentManagement
2	QBDT-Payroll-Service	https://devportal.intuit.com/app/dp/resource/7999536430416583774/incidentManagement

3	CEP	https://devportal.intuit.com/app/dp/resource/6343337667453040796/incidentManagement
4	Case 360	https://devportal.intuit.com/app/dp/resource/1587536460949765378/incidentManagement
5	EWS	https://devportal.intuit.com/app/dp/resource/8649180676662812116/incidentManagement
6	PTC	https://devportal.intuit.com/app/dp/resource/3315792817950450305/incidentManagement
7	QBDT Payroll Entitlement	https://devportal.intuit.com/app/dp/resource/6990730113881900139/incidentManagement (Not onboarded to A2i)
8	Workforce	https://devportal.intuit.com/app/dp/resource/2956630747670426490/incidentManagement
9	BRM	https://devportal.intuit.com/app/dp/resource/3893379470159715175/incidentManagement (Not onboarded to A2i)
10	Desktop Pay API	https://devportal.intuit.com/app/dp/resource/1730525749114265769/incidentManagement
11	Payroll DM	https://devportal.intuit.com/app/dp/resource/8866479358728757267/incidentManagement (Not onboarded to A2i since its not behind gateway)
12	Pubsub	https://devportal.intuit.com/app/dp/resource/651913638447486453/incidentManagement (Not onboarded to A2i since its not behind gateway)
13	EWS Service	https://devportal.intuit.com/app/dp/resource/4901059886671999792/incidentManagement
14	EWS-Rest-IKS Service	https://devportal.intuit.com/app/dp/resource/6270154173504901366/incidentManagement

Downtime Details

App	Tiptoe (PST)			Rollback (PST)		
	Downtime Start	Downtime End	Actual	Downtime Start	Downtime End	Actual
PSP	08/26/2023 22:34	08/26/2023 23:16	42 mins	08/27/2023 22:25	08/27/2023 23:11	46 mins
PAPI	08/26/2023 22:30	08/26/2023 23:25	55 mins	08/27/2023 22:21	08/27/2023 23:18	57 mins
Pubsub	08/26/2023 22:30	08/26/2023 23:24	54 mins	08/27/2023 22:21	08/26/2023 23:18	57 mins
RJF	08/26/2023 22:30	08/26/2023 23:25	55 mins	08/27/2023 22:21	08/26/2023 23:18	57 mins
Payroll DM	08/26/2023 22:30	08/26/2023 23:25	55 mins	08/27/2023 22:21	08/26/2023 23:18	57 mins

Downtime Summary

Downtime	Estimated	Actual	Tiptoe/Rollback Window		Comments
			Start Time (PST)	End Time (PST)	
Tiptoe	105 Mins	55 mins	08/26/2023 22:30	08/26/2023 23:25	Time Split Expected 1 hour 15 mins of Deploying all apps at full scale. 30 Mins for database side steps - Database Replication Lag is 0. - Stopping A B and creating B A replication tasks or vice versa in case of rollback.
Planned Rollback	120 Mins	57 mins	08/27/2023 22:21	08/26/2023 23:18	Time Split Expected - Database Replication Lag is 0. 15 Mins for Sanity.
Emergency Rollback (In case of issues during the Tiptoe window)	170 Mins	NOT EXECUTED	NA	NA	Time Split Expected 120 Mins for Planned Rollback 30 Mins Debugging and Replication Lag to settle down. 15 Mins of additional time for Sanity.

Learnings/Enhancements

S No	Learning	Action Item	Owner	Priority	Status	Comments
1	Baseline all the required queries.	- Capture all the queries needed baselining in desktop-payroll-database-engineering repo. - Baseline all the queries in Oracle Hash instance.	Goswami, Harshita	P0	COMPLETED	PSP-27768 - Getting issue details... STATUS
2	Understand and fix why the Adapter pod takes 25 mins to come healthy (min startup time - 8 vs max startup time - 25 mins)	Identify the root cause and fix.	N, Ramasubramani / Sur, Koustav	P2	OPEN	
3	Understand why One QBDT pod and 4 Adapter pods restarted during the Tiptoe activity	This is due to high CPU spike during start up, HPA triggered. Identify the root cause for CPU spike and fix.	Goswami, Harshita	P2	OPEN	
4	Look into the minimum database connection required for each adapter	- Keynote Adapter does not require any database connections. - VMP does not require any Read/Write database connections. - ADE Adapter can start with 1 minimum connection - DIS Adapter can start with 1 minimum connection Identify the root cause and fix.	Muthuranga m, Karthikeyan	P3	OPEN	
5	Dev Team should get the long running queries alert.	Route long running OLTP alert to #psp-notice-board channel. Email : psp-notice-board-aaaabidono7xnnna2mrc7n3kroe@intuit.org.slack.com	Choubey, Mayank	P0	OPEN	
6	AIA Gateway outgoing calls have 400 responses.	Identify the root cause and fix.	Singhal, Vikas	P3	OPEN	
7	AssistedUsageDataSyncProcessor batch job has some slow queries	Identify the slow queries and tune the queries in Oracle Range, Oracle Hash and Postgres.	Goswami, Harshita	P1	OPEN	
8	QBDT dashboard to capture exceptions, currently it returns 200 always.	Identify the root cause and fix.	Singhal, Vikas	P0	OPEN	
9	BulkWorkforceInvite Processor queries are slow.	Add Partition keys to the Queries to tune the queries.	Sur, Koustav / Saxena, Sakshi	P0	IN PROGRESS	
10	Opportunity to simplify Data Sanity.	Enhance Data Sanity scripts to complete faster.	Choubey, Mayank	P0	OPEN	
11	Performance dashboards for Pubsub/Payroll DM	Create new dashboards. Manually validate the performance during Tiptoe.	N, Ramasubramani	P1	OPEN	
12	QBDT Adapter receiving traffic while in maintenance	QBDT Adapter receiving traffic while in maintenance	N, Ramasubramani	P0	COMPLETED	Detailed analysis done. This is due to Node sanity for PSID 632004976.
13	Scale up and Scale Down job having Approve button before syncing to Argo	Scale up and Scale Down job having Approve button before syncing to Argo	Bhagat, Suyash	P2	OPEN	
14	Startup failures causing increased downtime.	Startup failures causing increased downtime. IDPS communication issues during startup.	Unknown User (mvenkatesh)	P2	OPEN	
15	PAPI and VMP not connecting to reader instances.	PAPI and VMP not connecting to reader instances.	Bhat, Chethan	P1	COMPLETED	There is no problem here. As soon as we receive traffic, we see connections.
16	SAP performance comparison	Dashboards for SAP workflows.	Goswami, Harshita	P0	COMPLETED	

17	RJF table privileges issues.	RJF report failed with Table of View does not exist error due to table privileges. How do we ensure all the objects, synonyms, roles and privileges are same across databases. Add this as pre tiptoe step in cutover wiki.	Choubey, Mayank	P0	OPEN	
18	All slow queries to be fixed.	All slow queries as per AWR Report to be fixed in Oracle Range and Oracle Hash	N. Ramasubramani / Muthuramgam, Karthikeyan	P3	OPEN	
19	Add Query Baselining in Release Wiki Template	Add Query Baselining in Release Wiki Template for any query introduction /modification	N. Ramasubramani	P0	COMPLETED	Added here PSP DB Review#OracleExitChecklist
20	Validate Global Indexes enablement in staging.	Enable Global Indexes on the following tables and validate this does not impact any queries, this is required for FMEA (When there is P0 workflow encounters performance issues) • psp_money_movement_transaction • psp_financial_transaction • psp_financial_trans_state • psp_entry_detail_record • psp_company_event • psp_company_event_detail • psp_company_event_email • psp_deduction • psp_tax	Sur, Koustav / Singhal, Vikas	P0	OPEN	
21	Enable all the batch jobs in Staging.	Enable all the batch jobs in Staging to test database stability.	Sur, Koustav	P0	COMPLETED	

Important Resources

Dev Portal

Application	Dev Portal Link
PSP	PSP Dev Portal
PAPI	PAPI Dev Portal
PubSub	Pubsub Dev Portal
RJF	RJF Dev Portal
PayrollDM	PayrollDM Dev Portal

JSS

Application	JSS Link
PSP Batch Jobs	PSP JSS View
RJF Reports	RJF JSS View

Splunk

Type	Link
Application	Splunk Application Production
Gateway	Splunk Gateway Production

ArgoCd

Environment	Application	Dev Portal Link
Production	PSP	Production PSP ArgoCd
	PAPI	Production PAPI ArgoCd
	Pubsub	Production Pubsub ArgoCd
	RJF	Production RJF ArgoCd
	PayrollDM	Production PayrollDM ArgoCd

Notification Email Addresses

i SBSEG-PD QBDT PSP AWS <SBSEG-PDQBDTSPAWS@intuit.com>; SBG PSP Services Decomp Scrum <SBGPSPServicesDecompScrum@intuit.com>; SBSEG SBG Payroll PSP Informed <SBGPayrollPSPInformed@intuit.com>; SBSEG-Iqbal-Extended <SBSEG-PDIqbalExtended@intuit.com>; SBSEG-PD DTPayrollExtended <SBSEG-PDDTPayrollExtended@intuit.com>; SBSEG-PD DTPayroll <SBSEG-PDDTPayroll@intuit.com> Kaura, Varun <Varun_Kaura@intuit.com>; Narlagalla, Praveen Goud <PraveenGoud_Narlagalla@intuit.com>; Swain, Amrit <Amrit_Swain@intuit.com>; Johnson, Anu <Anu_Johnson@intuit.com>, anuradha_kumari@intuit.com, rohan_rasam@intuit.com, narendraprasad_nanna@intuit.com, subulakshmi_nagarajan@intuit.com, rajeevkumar_singh@intuit.com, ketan_popat@intuit.com, Pahuja, Hitesh <Hitesh_Pahuja@intuit.com>, Huang, Alex <Alex_Huang@intuit.com>

Change Requests

Application	Tiptoe			Rollback		
	Change Request	Deployment Image	Status	Change Request	Deployment Image	Status
PSP	CHG2439062	master-45bbee7-eks	COMPLETED	CHG2439112	master-45bbee7-eks	COMPLETED
PAPI	CHG2439087	master-8d4f6ff	COMPLETED	CHG2439113	master-8d4f6ff	COMPLETED
Pubsub	CHG2439098	master-3bcc9d9	COMPLETED	CHG2439157	master-3bcc9d9	COMPLETED
RJF	CHG2439102	master-4b44388	COMPLETED	CHG2439159	master-4b44388	COMPLETED
PayrollDM	CHG2439095	master-ef5ddff	COMPLETED	CHG2439155	master-ef5ddff	COMPLETED

Runbook

Pre Tiptoe Steps

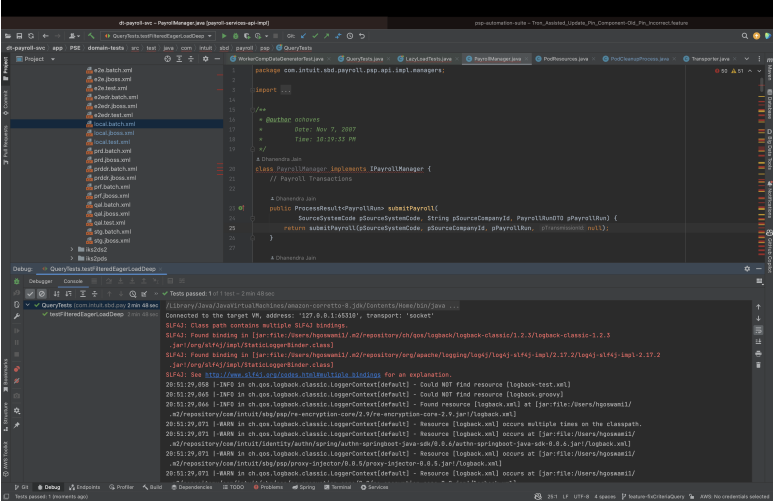
S No	Task	Steps	Due Date	Status	Owner
1	Partner Sign-off.	Refer NotificationEmailAddresses Section. Partner Contacts : Partner Team	22 Aug 2023	DONE	Nagarajan, Subulakshmi Nanna, Narendra Prasad
2	Inform partner teams slack channel.	Ask the stake holders to be prepared for the downtime. Inform following Slack Channels. #payroll-pd-releases, #ioc, #psp-rtb-priority, #sbg-payroll-prod, #mse-commerce-support, #cx3-production	23 Aug 2023	DONE	N, Ramasubramani
3	Send Release Freeze Email to Dev Team.	No Releases in PSP / PAPI / Pubsub / RJF / PayrollDM allowed in this interval.	20 Aug 2023	DONE	N, Ramasubramani
4	Care Sign-off	SAP can be validated in Staging Environment.	22 Aug 2023	DONE	Sur, Koustav

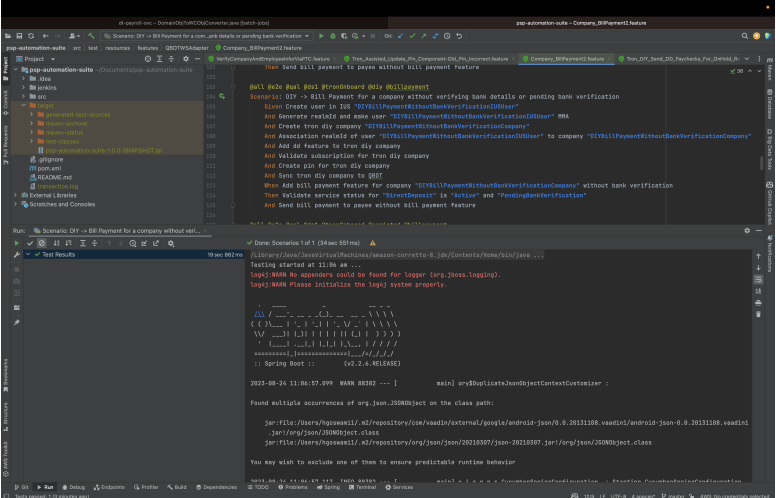
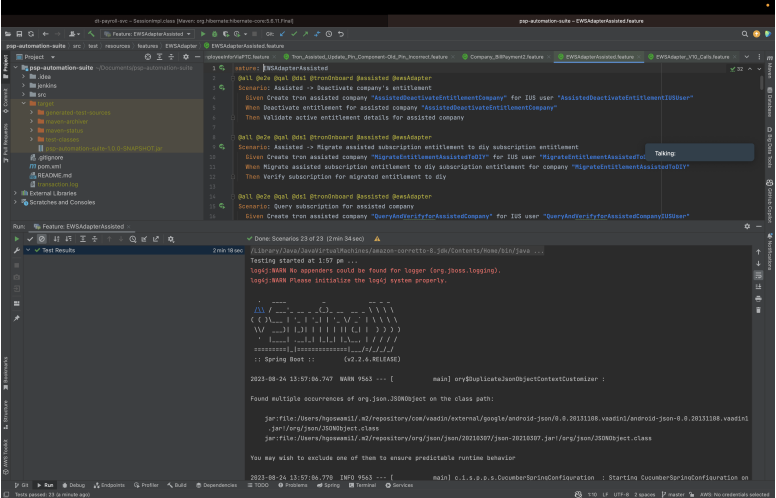
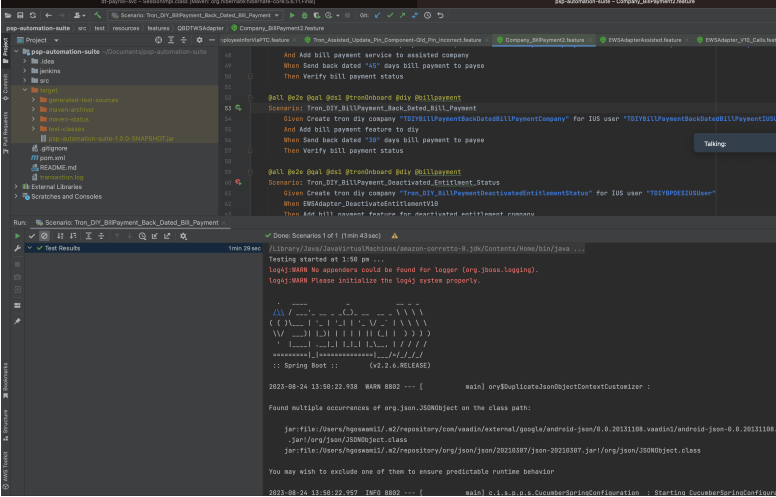
5	Inform Leadership Team on the Risk & Downtime	Inform Leadership Team on the Risk & Downtime.	24 Aug 2023	DONE	Nasim, Syed Iqbal												
6	Prepare QBDT Production Setup.	Prepare QBDT Production Setup.	23 Aug 2023	DONE	Swain, Amrit Holimath, Praveen Kumar												
7	Schema object count check	Count of objects are matching except known ones for partitioning.	25 Aug 2023	DONE	Choubey, Mayank												
8	Ensure no global indexes on partitioned tables	temp_drop_gl_indexes_for_val.sql	27 Aug 2023	DONE	Choubey, Mayank												
9	Time for Validating Latest data during the Tiptoe window	Unknown User (rrasam) / Rajoria, Himanshu to update expected numbers and veridata url and job names. <table><thead><tr><th>Activity</th><th>Job Name</th><th>Expected</th><th>Actual</th></tr></thead><tbody><tr><td>Tiptoe</td><td></td><td>9 days - ~3 hrs</td><td>9 days - ~3 hrs</td></tr><tr><td></td><td></td><td>24 hours - 2:5 hrs</td><td>24 hours - 2:5 hrs</td></tr></tbody></table>	Activity	Job Name	Expected	Actual	Tiptoe		9 days - ~3 hrs	9 days - ~3 hrs			24 hours - 2:5 hrs	24 hours - 2:5 hrs	23 Aug 2023	DONE	Rajoria, Himanshu / Unknown User (rrasam)
Activity	Job Name	Expected	Actual														
Tiptoe		9 days - ~3 hrs	9 days - ~3 hrs														
		24 hours - 2:5 hrs	24 hours - 2:5 hrs														
10	Decide strategy for Monthly & Quarterly Jobs.	Decide strategy for Monthly & Quarterly Jobs.	22 Aug 2023	NA	Swain, Amrit												
11	Make sure there is no IKS Cluster Upgrade during the Tiptoe window.	- Talk to MSaaS team and confirm. - Get the contacts of MSaaS team in case if we need assistance during tiptoe. - MSaaS Signoff - MSaaS Signoff Thread	22 Aug 2023	DONE	N, Ramasubramani												
12	Make sure there is no Splunk Maintenance during the Tiptoe window.	Slack Channel : #pi-tools-splunk-spt Splunk Signoff : Splunk Signoff Thread	22 Aug 2023	DONE	N, Ramasubramani												

13	We use PRs for the release. Make sure there is no Git maintenance during the Tiptoe window.	- Talk to Git team and confirm. - Git Signoff - Git Signoff Thread	22 Aug 2023	DONE	N, Ramasubramani																		
14	Capture 19 Aug 2023 - 20 Aug 2023 traffic for each app.	- Update AppsHealth section with the last week traffic. - Post the traffic and health numbers in tiptoe channel.	22 Aug 2023	DONE	Agarwal, Akshit Unknown User (akumari 3) Goswami, Harshita Singhal, Vikas																		
15	Make sure the last Master branch deployed did not have any deployment errors.	Look for errors in this dashboard <table><tr><th>App</th><th>Dashboard</th><th>Status</th></tr><tr><td>PSP</td><td>PSP Unknown Errors</td><td> NO ERRORS </td></tr><tr><td>PAPI</td><td>PAPI Unknown Errors</td><td> NO ERRORS </td></tr><tr><td>Pubsub</td><td>PubSub Unknown Errors</td><td> NO ERRORS </td></tr><tr><td>RJF</td><td>RJF Unknown Errors</td><td> NO ERRORS </td></tr><tr><td>PayrollDM</td><td>PayrollDM Unknown Errors</td><td> NO ERRORS </td></tr></table>	App	Dashboard	Status	PSP	PSP Unknown Errors	NO ERRORS	PAPI	PAPI Unknown Errors	NO ERRORS	Pubsub	PubSub Unknown Errors	NO ERRORS	RJF	RJF Unknown Errors	NO ERRORS	PayrollDM	PayrollDM Unknown Errors	NO ERRORS	22 Aug 2023	DONE	Sur, Koustav
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Pubsub	PubSub Unknown Errors	NO ERRORS																					
RJF	RJF Unknown Errors	NO ERRORS																					
PayrollDM	PayrollDM Unknown Errors	NO ERRORS																					
16	Meet to decide GO/No GO	GOOD TO GO	23 Aug 2023	DONE	N, Ramasubramani Muthuram, KarthikeyanNasim, Syed Iqbal Choubey, Mayank Bhat, Chethan																		
17	Send Downtime Update to stakeholders via Email.	Refer NotificationEmailAddresses Section. Ask the stakeholders to be prepared for the downtime.	23 Aug 2023	DONE	N, Ramasubramani																		
18	Reassign PagerDuty.	1. Update PagerDuty Schedule for the Tiptoe window. 2. Update PagerDuty Schedule for the next week after the Tiptoe. Pager Duty Schedules QBPY - DPS : EMS OPS RNO Primary SBG - QBPY - DPS : DevOps PSP (IDC)	25 Aug 2023	DONE	N, Ramasubramani																		

19	Note down all the PSP JSS Jobs in the same window.	Link : PSP JSS View Update Failed/MissedBatchJobs Section.	23 Aug 2023	DONE	Hoolimat h, Praveenkumar
20	Note down all the RJF JSS Jobs in the same window.	Link : RJF JSS View Update Failed/MissedBatchJobs Section.	23 Aug 2023	DONE	Hoolimat h, Praveenkumar
21	Update IDPS Oracle Hash Credentials. - data Accession. url - data Accession. user name - data Accession. password - data Accession. read Only. url	Note: for tiptoe, Oracle C has only RW instance. Readers will switch back to Oracle A. We should divert traffic to RW before switching. Steps: Pre-requisite: install StashCopy .pem file to /tmp folder. eiamcli login - Add Secrets from Command Line. <code>sh Oracle-Range-A.sh -a</code> <code>sh Oracle-Hash-B.sh -a</code> <code>sh Oracle-Range-C.sh -a</code> - Validate Secrets from Command Line. <code>sh Oracle-Range-A.sh -v</code> <code>sh Oracle-Hash-B.sh -v</code> <code>sh Oracle-Range-C.sh -v</code> - Validate from Developer Portal	23 Aug 2023	DONE	Bansal, Jatin

	5. data Access. connection. read Only. user name 6. data Access. connection. read Only. password																						
22	Update CR and Image Details	Create Change Requests & decide images for Tiptoe and Rollback. Refer ChangeRequests Section	23 Aug 2023	DONE	N, Ramasubramani																		
23	Bring down staging and middle ware apps.	- Bring down staging and middleware apps. - Ensure no database connections from Middleware and Staging. - Bringdown the databases.	25 Aug 2023	DONE	Agarwal, Akshit Choubey, Mayank																		
24	Ensure there is no OOS or any data issue between Oracle Range and Oracle Hash	- Veridata reports should not have any OOS. - Ensure latest/last 9 days and 1 day compare is green. Veridata Console - Job Name: monolith_rhab_incr_9days, monolith_rhab_incr_1day	23 Aug 2023	DONE	Unknown User (rrasam) / Rajoria, Himanshu																		
25	Run Sanity for PSP /PAPI /Pubsub/RJF. Execute the sanity at 9AM on the Tiptoe day.	Execute below Sanity Jobs to ensure jobs working fine as expected. <table><tr><th>App</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PSP</td><td>PSP Sanity</td><td> DONE </td></tr><tr><td>PAPI</td><td>PAPI Sanity</td><td> DONE </td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td><td> DONE </td></tr><tr><td>RJF</td><td>RJF Sanity</td><td> DONE </td></tr><tr><td>PayrollDM</td><td>PayrollDM Sanity</td><td> DONE </td></tr></table>	App	Sanity Link	Status	PSP	PSP Sanity	DONE	PAPI	PAPI Sanity	DONE	Pubsub	Pubsub Sanity	DONE	RJF	RJF Sanity	DONE	PayrollDM	PayrollDM Sanity	DONE	26 Aug 2023	DONE	Bansal, Jatin
App	Sanity Link	Status																					
PSP	PSP Sanity	DONE																					
PAPI	PAPI Sanity	DONE																					
Pubsub	Pubsub Sanity	DONE																					
RJF	RJF Sanity	DONE																					
PayrollDM	PayrollDM Sanity	DONE																					

26	Validate PSP Node Level Sanity.	Validate Node level Sanity and Dashboard Node Level Sanity Dashboard.	24 Aug 2023	DONE	Bansal, Jatin
27	Update Security Group to add Production Connection	Update Security Group to add Production Connection	23 Aug 2023	DONE	Choubey, Mayank
28	Integration Test Suite	https://build.intuit.com/intuit-desktop-payroll/job/iks-intg/4107/ only one additional test failed . attached evidence 	24 Aug 2023	DONE	Sur, Koustav / N, Ramasubramani

29	Automation Test Suite	https://build.intuit.com/intuit-desktop-payroll/job/payroll-automation-suite-e2e/2396/cucumber-html-reports/report-feature_55_3646969197.html	24 Aug 2023	DONE	Sur, Koustav
		failed tests ran locally attached evidence of intersection of above two test cases			
					
					
					
30	Run data sanity checks for all main tables.	sanity-table-rowcount-check.sql	23 Aug 2023	DONE	Unknown User (rrasam)

31	Sequences Validation between Oracle range and oracle hash	Compare last_number for sequences between Oracle range and oracle hash. It should be 0 or greater than source.	27 Aug 2023	DONE	Choubey, Mayank
32	Prepare Stash Commands to replace Oracle Range database credentials with Oracle Hash database credentials	Note: for tiptoe, Oracle C has only RW instance. Readers will switch back to Oracle A. We should divert traffic to RW before switching. • Steps: • Pre-requisite: install Stash • Copy .pem file to /tmp folder. • eiamcli login • Add Secrets from Command Line. <i>sh Oracle-Range-A.sh -a</i> <i>sh Oracle-Hash-B.sh -a</i> <i>sh Oracle-Range-C.sh -a</i> • Validate Secrets from Command Line. <i>sh Oracle-Range-A.sh -v</i> <i>sh Oracle-Hash-B.sh -v</i> <i>sh Oracle-Range-C.sh -v</i> • Validate from Developer Portal	23 Aug 2023	DONE	Bansal, Jatin
33	Create Global indexes for Range Partitioned tables.	Create Global indexes for the following tables. • psp_entitlement_message • psp_paycheck_usage	24 Aug 2023	DONE	Choubey, Mayank
34	Team to validate Database connectivity for A/B/C Prod instances.	• Choubey, Mayank to create a read-only user on A/B/C instances. • Team to validate Database connectivity for A/B/C Prod instances.	24 Aug 2023	DONE	Choubey, Mayank Johnson, Anu
35	Update PayrollDM Config to remove hard coded DB URL / Username / Password.	Update PayrollDM Config to remove hard coded DB URL / Username / Password.	24 Aug 2023	DONE	Bhagat, Suyash
36	Prepare for bad performing queries	• Baseline the queries - Choubey, Mayank Popat, Ketan Gunvantrai • Checkin the slow queries here - Goswami, Harshita Singhal, Vikas Bansal, Jatin	25 Aug 2023	DONE	Choubey, Mayank Popat, Ketan Gunvantrai
37	Update Maintenance Window	Update Maintenance Window. Refer Maintenance Windows section. • Sunday (10AM - 1PM) IST • Monday (10AM - 1PM) IST Refer below threads and confirm they snoozed their asset as well. • MSE Team • CX3 Team	25 Aug 2023	DONE	Bhagat, Suyash

TipToe Steps

S No	Task	Steps	Status	Owner	Remarks														
1	Update Maintenance Window	Update Maintenance Window. Refer Maintenance Windows section. - Sunday (10AM - 1PM) IST - Monday (10AM - 1PM) IST Refer below threads and confirm they snoozed their asset as well. MSE Team CX3 Team	DONE	Bhagat, Suyash															
2	Data Parity	Ensure latest 1 day compare and sanity check output is green.	DONE	Unknown User (rrasam) Rajoria, Himanshu															
3	Send Downtime Update to stake holders via Email.	CR, Apps & Downtime details to be shared. Refer ChangeRequests Section. Refer NotificationEmailAddresses Section.	DONE	N, Ramasubramani															
4	Inform partner teams slack channel.	Ask the stake holders to be prepared for the downtime. CR, Apps & Downtime details to be shared. Refer ChangeRequests Section. Inform following Slack Channels. #payroll-pd-releases, #ioc, #psp-rtb-priority, #sbg-payroll-prod, #mse-commerce-support, #cx3-production	DONE	N, Ramasubramani															
5	Ensure there are no critical jobs running.	- Navigate to JSS View Link. - Wait for critical jobs to complete.	DONE	Hoolimath, Praveenkumar															
6	Bring down below apps. - PayrollDM - Pubsub - RJF - PAPI	Bring down Pubsub, PAPI, RJF and PayrollDM. Zero Pods With Maintenance Page	DONE	Bansal, Jatin															
7	Bring down PSP along with Maintenance Page	Bring down PSP with Maintenance Page. Zero Pods PSP With Maintenance Page	DONE	Bansal, Jatin															
8	Ensure all apps are down.	Hit following URLs for each application. <table><tr><th>App</th><th>Health Check URL</th></tr><tr><td>PSP</td><td>sh PSPHealthCheck.sh prd</td></tr><tr><td>SAP</td><td>SAP</td></tr><tr><td>PAPI</td><td>PAPI Health Check</td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td></tr><tr><td>RJF</td><td>RJF Health Check</td></tr><tr><td>PayrollDM</td><td>PayrollDM Health Check</td></tr></table>	App	Health Check URL	PSP	sh PSPHealthCheck.sh prd	SAP	SAP	PAPI	PAPI Health Check	Pubsub	Pubsub Sanity	RJF	RJF Health Check	PayrollDM	PayrollDM Health Check	DONE	Goswami, Harshita	
App	Health Check URL																		
PSP	sh PSPHealthCheck.sh prd																		
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9	Update the downtime details section for all the apps.	Update DowntimeDetails Section.	DONE	Muthurangam, Karthikeyan															
10	Note down All the failed RJF/PSP BatchJobs.	Update Failed/MissedBatchJobs Section.	DONE	Hoolimath, Praveenkumar															

11	Update Production to point to Oracle Hash(B) Database.	Update Read Write / VMP / RJF Read Instances (3 instances in total). • Pre-requisite: install Stash • Copy .pem file to /tmp folder. • eiamcli login • Update IDPS Credentials (Database URL/Username /Password) <i>sh Oracle-Hash-B.sh -a</i> • Validate IDPS Credentials (Database URL/Username /Password) <i>sh Oracle-Hash-B.sh -v</i> Bansal, Jatin / Sur, Koustav to update	DONE	Sur, Koustav	
12	Lock Oracle Range (A instance) users	Lock Oracle Range Users.	DONE	Choubey, Mayank	
13	Ensure there are no active connections to the pre-production Range database. (Kill connections if needed)	• Wait for Inflight requests/Running Batch Jobs to complete - time boxed to 2 mins. • There should not be any ACTIVE connections to Oracle Range Database. • No active connections from PSP / PAPI / Pubsub / RJF / PayrollDM. chk-kill-db-connections.sql	DONE	Choubey, Mayank	
14	Add new extract for A->P	login to ggpspuwp01.sbg-psp-prod.a.intuit.com <pre>dblogin useridalias ggsource obey ./diroby/create_epspnwp1.oby register extract epspnwp1 database info PPSPEP05 detail stop PPSPEP05 info PPSPEP05 detail</pre>	DONE	Choubey, Mayank	
15	Monitor Oracle Range Oracle Hash Replication Lag	Wait till Oracle Range (A) Oracle Hash(B) Replication lag to zero. Keep eye on BC1 GG replication.	DONE	Choubey, Mayank	
16	GG Replication from Oracle Hash (B) Oracle Range(A)	Ensure Extract (B) stats should not have any table extract stats stats <extract_name>	DONE	Choubey, Mayank	
17	Unlock users in Oracle Hash (B)	application-accounts-unlock.sql	DONE	Choubey, Mayank	
18	Restart Extract on Range db (A)	• To ensure nothing gets extracted after user lock on Range (A) • stop EPSWP01 • start EPSWP01	DONE	Choubey, Mayank	
19	Bring up PSP Pods with Maintenance.	Bring up PSP with Maintenance page. Scale up PSP Pods With Maintenance Page. Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd	DONE	Bansal, Jatin	
20	Monitor if PSP connecting to Oracle Hash Database	Monitor Database Connection Dashboard and validate.	DONE	Rajvanshi, Venkat	
21	Ensure there are no errors in the startup logs.	Look for errors in this dashboard Unknown Errors	DONE	Bhagat, Suyash	
22	Check the Node level Sanity Logs.	Node Level Sanity Dashboard - Check if all the pods passed sanity tests.	DONE	Rajvanshi, Venkat	
23	Monitor DB Connections.	Monitor db connetions	DONE	Choubey, Mayank	

24	Monitor Overall database /replication health vi CW dashboard	Oracle Hash Database Performance Monitor GoldenGate Replication Dashboard	DONE	Choubey, Mayank																
25	Bring up PSP pods without Maintenance.	Bring up PSP without Maintenance page. Scale up PSP Pods Without Maintenance Page. Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd	DONE	Bansal, Jatin																
26	Run the PSP Sanity.	Execute below Sanity Jobs to ensure jobs working fine as expected. <table><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PSP</td><td>PSP Sanity</td><td> SUCCESS </td></tr><tr><td>SAP</td><td>SAP Sanity</td><td> SUCCESS </td></tr></table>	Application	Sanity Link	Status	PSP	PSP Sanity	SUCCESS	SAP	SAP Sanity	SUCCESS	DONE	Goswami, Harshita							
Application	Sanity Link	Status																		
PSP	PSP Sanity	SUCCESS																		
SAP	SAP Sanity	SUCCESS																		
27	Run replication sanity (count based table records validation)	sanity-table-rowcount-check.sql	DONE	Unknown User (rrasam)																
28	Update the downtime details section for PSP.	Update DowntimeDetails Section.	DONE	Muthurangam , Karthikeyan																
29	Monitor DB Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	DONE	Choubey, Mayank Popt, Ketan Gunvantrai																
30	Bring up all pods for below services 1. PAPI 2. RJF 3. Pubsub 4. PayrollDM	Bring up Pubsub, PAPI, RJF and PayrollDM apps without maintenance page. Bring up Apps Without Maintenance Page Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd	DONE	Bansal, Jatin																
31	Monitor if all apps connecting to Oracle Hash Database	Monitor Database Connection Dashboard and validate.	DONE	Rajvanshi, Venkat																
32	Ensure there are no errors in the startup logs.	Monitor and Report unknown errors. <table><tr><th>Application</th><th>Dashboard</th></tr><tr><td>PAPI</td><td>PAPI Unknown Errors</td></tr><tr><td>Pubsub</td><td>PubSub Unknown Errors</td></tr><tr><td>RJF</td><td>RJF Unknown Errors</td></tr><tr><td>PayrollDM</td><td>Payroll DM Unknown Errors</td></tr></table>	Application	Dashboard	PAPI	PAPI Unknown Errors	Pubsub	PubSub Unknown Errors	RJF	RJF Unknown Errors	PayrollDM	Payroll DM Unknown Errors	DONE	Bhagat, Suyash						
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PAPI	PAPI Unknown Errors																			
Pubsub	PubSub Unknown Errors																			
RJF	RJF Unknown Errors																			
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33	Run PAPI/Pubsub/RJF/PayrollDM Sanity.	Execute below Sanity Suites. <table><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PAPI</td><td>PAPI Sanity</td><td> SUCCESS </td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td><td> SUCCESS </td></tr><tr><td>RJF</td><td>RJF Sanity</td><td> SUCCESS </td></tr><tr><td>PayrollDM</td><td>PayrollDM Sanity</td><td> SUCCESS </td></tr></table>	Application	Sanity Link	Status	PAPI	PAPI Sanity	SUCCESS	Pubsub	Pubsub Sanity	SUCCESS	RJF	RJF Sanity	SUCCESS	PayrollDM	PayrollDM Sanity	SUCCESS	DONE	Goswami, Harshita	
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PAPI	PAPI Sanity	SUCCESS																		
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RJF	RJF Sanity	SUCCESS																		
PayrollDM	PayrollDM Sanity	SUCCESS																		
34	Ensure we see DB Connections from Pubsub / PAPI / RJF / PayrollDM.	Monitor db connetions	DONE	Choubey, Mayank																
35	Update the downtime details section for all the apps.	Update DowntimeDetails Section.	DONE	Muthurangam , Karthikeyan																
36	Run PSP & RJF failed jobs and missed Jobs.	Refer Failed/MissedBatchJobs Section. Run LedgerBalance Job if Missed.	DONE	Hoolimath, Praveenkumar																

37	Monitor all the apps and databases.	Refer Monitoring Section.	DONE	Everyone	
38	Ensure replication is running without any issues & monitor the lag	GoldenGate Replication Dashboard	DONE	Choubey, Mayank	
39	Review & Take action on all the pending Alerts.	Pending Alerts - Critical Alerts PD Pending Alerts - RNO PD	DONE	N, Ramasubramani	
40	Run Payroll From Production Test company	IKS QBDT PSP Adapters	DONE	Swain, Amrit	
41	Monitor RJF & PSP Batch Jobs.	- Update JSS Jobs Schedule completed jobs status every 2hours.(validate job using sql queries) - Update RJF Jobs Schedule completed jobs status every 2hours.(validate job with splunk)	DONE	Hoolimath, Praveenkumar	

Post TipToe Steps

S No	Task	Steps	Status	Owner	Remarks
1	Send Downtime Update to stake holders via Email.	Refer NotificationEmailAddresses Section.	DONE	N, Ramasubramani	
2	Offline Monitoring for all the Apps.	Refer Monitoring Section. Update once in every 2 hours in tiptoe slack channel.	DONE	All	
3	Review & Take action on all the pending Alerts.	Pending Alerts - Critical Alerts PD Pending Alerts - RNO PD	DONE	Hoolimath, Praveenkumar	
4	Review & Close all pending CRs.	Refer ChangeRequests Section.	DONE	N, Ramasubramani	
5	Inform partner teams slack channel.	Inform following Slack Channels. #payroll-pd-releases, #ioc, #psp-rtb-priority, #sbg-payroll-prod, #mse-commerce-support, #cx3-production	DONE	N, Ramasubramani	
6	Monitor Critical Batch Jobs.	Monitor Critical Batch Jobs (Required Only for the Cutover) Refer CriticalBatchJobsMonitoring Section.	DONE	Hoolimath, Praveenkumar	
7	Monitor database Stability, Availability & Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	DONE	Choubey, Mayank (Primary) Popat, Ketan Gunvantrai (Escalation)	
8	Monitor replication from Oracle Hash to Oracle Range	- Monitor DMS replication lag from Oracle Hash to Oracle Range - Keep running sanity counts. - Ensure Veridata validation is green from Oracle Hash to Oracle Range	DONE	Narlagalla, Praveen Goud	
9	Schedule Veridata to validate Delta Data.	Schedule Veridata to validate Delta Data. Veridata Console Link Veridata Job Name: monolith_hrba_incr_1day	DONE	Unknown User (rrasam) Rajoria, Himanshu	
10	Apply Repair Scripts if applicable.	Apply Repair Scripts if applicable.	DONE	Choubey, Mayank	

Rollback Decision and Action Plan

Issue Category	Scenario	Action Item
Application Functionality	Any of the following issues. - P0/P1 customers workflow impacted. - P0/P1 agents workflow impacted. - P0/P1 Batchjobs impacted.	Rollback and Restore application/service.

	P2/P3 customers/agents/BatchJobs issue.	No Rollback. Fix and Move Forward.
Application Performance	Any of the following issues. - P0/P1 customers workflow impacted. - P0/P1 agents workflow impacted. - P0/P1 Batchjobs impacted.	1. Try below steps if this works. - Flush the current plan. - Baseline the plan. 2. If nothing works, if performance degrade is in acceptable range, move forward. 3. If nothing works, if performance degrade is not in acceptable range, Rollback and Restore.
	P2/P3 customers/agents/BatchJobs issue.	Fix and move forward. Try below steps. - Flush the current plan. - Baseline the plan. Stay even if performance is degraded. Pause the jobs if required.
VMP/Equifax	VMP and Equifax performance degraded.	Switch VMP and Equifax traffic from Reader to Read/Write instance.
Replication	Replication task is running into error and replication is broken due to following scenarios.	Replication broken in A and C. Fix and move forward. No rollback is possible here.
Data parity	Data Parity fails	Fix and Forward. Massive Out of Sync in 1 instance. Generate OOS fix script and fix OOS.

Rollback Steps

	Task	Steps	Status	Owner	Remarks
1	Update Maintenance Window	Update Maintenance Window. Refer Maintenance Windows section. - Sunday (10AM - 1PM) IST - Monday (10AM - 1PM) IST Refer below threads and confirm they snoozed their asset as well. MSE Team CX3 Team	DONE	Bhagat, Suyash	
2	Ensure Data comparison is clean, run data-fix if applicable.	Unknown User (rrasam) to update veridata url and job name	DONE	Unknown User (rrasam) Rajoria, Himanshu Choubey, Mayank	
3	Ensure B A Replication lag is 0. [Skip this step for Emergency Rollback]	Monitor Oracle Hash --> Oracle Range Replication lag.	DONE	Choubey, Mayank	
4	[Optional] Consider going to C , if replication is broken from BA or lag is too high only for B A. [Skip this step for Emergency Rollback]	- Decide based on Replication lag & Data Parity check. - Stop BA all replication tasks	NA	Choubey, Mayank	
5	Peer review #1 and #2 [Skip this step for Emergency Rollback]	Decide based on Replication lag & Data Parity check.	NA	Choubey, Mayank	
6	Note down all the PSP JSS Jobs in the same window.	Link : PSP JSS View Update Failed/MissedBatchJobs Section.	DONE	Sur, Koustav	
7	Note down all the RJF JSS Jobs in the same window.	Link : RJF JSS View Update Failed/MissedBatchJobs Section.	DONE	Sur, Koustav	

8	Inform slack channel about the downtime.	Ask the stake holders to be prepared for the downtime. CR, Apps & Downtime details to be shared. Refer ChangeRequests Section.Inform following Slack Channels. #payroll-pd-releases, #ioc, #psp-rtb-priority, #sbg-payroll-prod, #mse-commerce-support, #cx3-production	DONE	N, Ramasubramani															
9	Send Downtime Update to stake holders via Email.	CR, Apps & Downtime details to be shared. Refer ChangeRequests Section. Refer NotificationEmailAddresses Section.	DONE	N, Ramasubramani															
10	Ensure there are no critical batch jobs/RJF jobs running.	Refer JSS View Link. Wait for critical jobs to complete.	DONE	Hoolimath, Praveenkumar															
11	Bring down below pods 1. PayrollDM 2. Pubsub 3. RJF 4. PAPI	Bring down Pubsub, PAPI (With Maintenance Page), RJF and PayrollDM. Zero Pods With Maintenance Page	DONE	Agarwal, Akshit															
12	Bring down PSP along with Maintenance Page	Bring Down PSP with Maintenance Page. Zero Pods PSP With Maintenance Page.	DONE	Agarwal, Akshit															
13	Ensure all apps are down.	Hit Following URLs and make sure all apps are down. <table><tr><th>Application</th><th>Health Check</th></tr><tr><td>PSP</td><td>sh PSPHealthCheck.sh prd</td></tr><tr><td>SAP</td><td>SAP</td></tr><tr><td>PAPI</td><td>PAPI Health Check</td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td></tr><tr><td>RJF</td><td>RJF Health Check</td></tr><tr><td>PayrollDM</td><td>PayrollDM Health Check</td></tr></table>	Application	Health Check	PSP	sh PSPHealthCheck.sh prd	SAP	SAP	PAPI	PAPI Health Check	Pubsub	Pubsub Sanity	RJF	RJF Health Check	PayrollDM	PayrollDM Health Check	DONE	Rajvanshi, Venkat	
Application	Health Check																		
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14	Update the downtime details section for all the apps.	Update DowntimeDetails Section.	DONE	Muthurangam, Karthikeyan															
15	Note down All the failed RJF/PSP BatchJobs.	Update Failed/MissedBatchJobs Section.	DONE	Hoolimath, Praveenkumar															
16	Update Production to point to Oracle Range(A or C) Database.	Choubey, Mayank to decide if the instance is A or C. Update Read Write / VMP / RJF Read Instances (3 instances in total). • Pre-requisite: install Stash • Copy .pem file to /tmp folder. • eiamcli login • Update IDPS Credentials (Database URL/Username /Password) <code>sh Oracle-Range-A.sh -a</code> OR <code>sh Oracle-Range-C.sh -a</code> • Validate IDPS Credentials (Database URL/Username /Password) <code>sh Oracle-Range-A.sh -v</code> OR <code>sh Oracle-Range-C.sh -v</code> Note: for tiptoe, Oracle C has only RW instance. Readers will switch back to Oracle A. We should divert traffic to RW before switching. Bansal, Jatin / Sur, Koustav to update	DONE	Sur, Koustav															
17	Lock Oracle Hash Users on B	application-accounts-lock.sql	DONE	Choubey, Mayank															
18	Ensure there are no active connections to the Oracle Hash Database. Kill sessions (if needed)	• Wait for Inflight requests/Running Batch Jobs to complete. • There should not be any connections to Oracle Hash Database. • No active connections from PSP / PAPI / Pubsub / RJF / PayrollDM. chk-kill-db-connections.sql	DONE	Choubey, Mayank															
19	Monitor Oracle Hash Oracle Range Replication Lag	Wait till Oracle Hash --> Oracle Range Replication lag to zero. Keep eye on BC1 GG replication.	DONE	Choubey, Mayank															

20	Validate Delta Data. Apply Repair Scripts if applicable.	Execute quick sanity check script. sanity-table-rowcount-check.sql Apply Repair Scripts if applicable.	DONE	Unknown User (rrasam)										
21	Unlock Oracle Range Users	- Unlock A or C based on the decision. - Unlock Oracle Range Users. application-accounts-unlock.sql	DONE	Choubey, Mayank										
22	Restart Extract for Hash db (B)	To ensure no transactions on B and nothing gets extracted from Hash db (B) stop EPSPHWP2 start EPSPHWP2	DONE	Choubey, Mayank										
23	Execute this step if we decide to come back to C. Update IDPS credentials for Oracle.	Update IDPS credentials for Oracle Range (A or C). Pre-requisite: install Stash - Copy .pem file to /tmp folder. - eiamcli login - Update IDPS Credentials (Database URL/Username /Password) sh Oracle-Range-A.sh -a OR sh Oracle-Range-C.sh -a - Validate IDPS Credentials (Database URL/Username /Password) sh Oracle-Range-A.sh -v OR sh Oracle-Range-C.sh -v Note: for tiptoe, Oracle C has only RW instance. Readers will switch back to Oracle A. We should divert traffic to RW before switching.	DONE	Sur, Koustav										
24	Bring up PSP pods with Maintenance.	Bring up PSP with Maintenance page. Scale up PSP Pods With Maintenance Page. Ensure correct image is deployed. Dashboard : Validate from DT Payroll Pod Info Script : imageDetails.sh prd	DONE	Agarwal, Akshit										
25	Monitor if PSP connecting to Oracle Range Database	Monitor Database Connection Dashboard and validate.	DONE	Goswami, Harshita										
26	Ensure there are no errors in the startup logs.	Look for errors in this dashboard Unknown Errors	DONE	Bhagat, Suyash										
27	Check the Node level Sanity Logs.	Node Level Sanity Dashboard - Check if all the pods passed sanity tests.	DONE	Goswami, Harshita										
28	Monitor DB Connections on Oracle Range.	Monitor db connetions	DONE	Choubey, Mayank										
29	Monitor Overall database /replication health vi CW dashboard	Oracle Hash Database Performance Monitor GoldenGate Replication Dashboard	DONE	Choubey, Mayank										
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31	Run the PSP Sanity.	Execute below Sanity Jobs to ensure jobs working fine as expected. <table><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PSP</td><td>PSP Sanity</td><td> SUCCESS </td></tr><tr><td>SAP</td><td>SAP Sanity</td><td> SUCCESS </td></tr></table>	Application	Sanity Link	Status	PSP	PSP Sanity	SUCCESS	SAP	SAP Sanity	SUCCESS	DONE	Rajvanshi, Venkat	
Application	Sanity Link	Status												
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32	Monitor DB Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	DONE	Choubey, Mayank / Popat, Ketan Gunvantrai										

33	Bring up all pods for below services 1. PAPI 2. RJF 3. Pubsub 4. PayrollDM	Bring up Pubsub, PAPI, RJF and PayrollDM apps without maintenance page. Bring up Apps Without Maintenance Page Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd	DONE	Agarwal, Akshit																
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36	Run PAPI / Pubsub / RJF / PayrollDM Sanity.	Execute below sanity suites. <table><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PAPI</td><td>PAPI Sanity</td><td> SUCCESS </td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td><td> SUCCESS </td></tr><tr><td>RJF</td><td>RJF Sanity</td><td> SUCCESS </td></tr><tr><td>PayrollDM</td><td>PayrollDM Sanity</td><td> SUCCESS </td></tr></table>	Application	Sanity Link	Status	PAPI	PAPI Sanity	SUCCESS	Pubsub	Pubsub Sanity	SUCCESS	RJF	RJF Sanity	SUCCESS	PayrollDM	PayrollDM Sanity	SUCCESS	DONE	Rajvanshi, Venkat	
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38	Monitor all the apps and databases.	Refer Monitoring Section.	DONE	All																
39	Ensure replication is running without any issues & monitor the lag	Monitor Replication Lag.	DONE	Narlagalla, Praveen Goud																
40	Run Payroll From Production Test company	IKS QBDT PSP Adapters	DONE	Swain, Amrit																

Post Rollback Steps

S No	Task	Steps	Status	Owner	Remarks
1	Send Downtime Update to stake holders via Email.	Refer NotificationEmailAddresses Section.	DONE	N, Ramasubramani	
2	Offline Monitoring for all Apps for 2 hours.	Refer Monitoring Section.	DONE	All	
3	Review & Take action on all the pending Alerts.	Pending Alerts - Critical Alerts PD Pending Alerts - RNO PD	DONE	Kaura, Varun	
4	Review & Close all pending CRs.	Refer ChangeRequests Section.	DONE	N, Ramasubramani	
5	Inform partner teams slack channel.	Ask the stake holders to be prepared for the downtime. Inform following Slack Channels. #payroll-pd-releases, #ioc, #psp-rtb-priority, #sbg-payroll-prod, #mse-commerce-support, #cx3-production	DONE	N, Ramasubramani	
7	Monitor Critical Batch Jobs.	Monitor Critical Batch Jobs (Required Only for the Cutover) Refer CriticalBatchJobsMonitoring Section.	DONE	Hoolimath, Praveenkumar	

8	Monitor database Stability, Availability & Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	DONE	Choubey, Mayank (Primary) Popat, Ketan Gunvantrai (Escalation)	
9	Monitor replication from Oracle Range to Oracle Hash	Ensure no replication issue	DONE	Choubey, Mayank	
10	Trigger Veridata to validate Delta Data.	Trigger Veridata to validate Delta Data. Veridata Console Link : monolith_rhab_incr_1day	DONE	Unknown User (rrasam) Rajoria, Himanshu	

Monitoring

Apps Health

- Tiptoe Hourly Monitoring captured here [Monolith Tip-Toe \(Oracle Hash\) Monitoring](#)
- Health & Traffic summary shared in the tiptoe slack channel.

Application / Database / Infrastructure Category	Owner	Health			
		Previous Week	Previous Week	Tiptoe	Rollback
		19 Aug 2023	20 Aug 2023	26 Aug 2023	27 Aug 2023
QBDT Adapter	Singhal, Vikas	HEALTHY	HEALTHY	HEALTHY	HEALTHY
QBDTWS Adapter	Singhal, Vikas	HEALTHY	HEALTHY	HEALTHY	HEALTHY
DIS Adapter	Singhal, Vikas	HEALTHY	HEALTHY	HEALTHY	HEALTHY
ADE Adapter	Goswami, Harshita	HEALTHY	HEALTHY	HEALTHY	HEALTHY
CDM Adapter	Goswami, Harshita	HEALTHY	HEALTHY	HEALTHY	HEALTHY
SAP Adapter	Goswami, Harshita	HEALTHY	HEALTHY	HEALTHY	HEALTHY
BRM Adapter	Unknown User (akumari3)	HEALTHY	HEALTHY	HEALTHY	HEALTHY
PTC Adapter	Unknown User (akumari3)	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Critical Batch Jobs	Hoolimath, Praveenkumar	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Batch Jobs during Activity	Hoolimath, Praveenkumar	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Infrastructure Monitoring	Joshi, Vishwanath	HEALTHY	HEALTHY	HEALTHY	HEALTHY
PAPI	Agarwal, Akshit	HEALTHY	HEALTHY	HEALTHY	HEALTHY
RJF	Sur, Koustav	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Pubsub	Agarwal, Akshit	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Payroll DM	Agarwal, Akshit	HEALTHY	HEALTHY	HEALTHY	HEALTHY
DB Queries Performance	Sur, Koustav / Rajvanshi, Venkat	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Database	Choubey, Mayank	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Veridata	Choubey, Mayank	HEALTHY	HEALTHY	HEALTHY	HEALTHY
ArgoCd	Joshi, Vishwanath	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Unknown Errors	Bhagat, Suyash	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Workflows Performance	Sur, Koustav / Rajvanshi, Venkat	HEALTHY	HEALTHY	HEALTHY	HEALTHY

PSP

S. No	Dashboard	Link	Comments
1	PSP Batch Jobs 1. Successful Jobs 2. Failed Jobs	PSP Batch Jobs	Update JSS Jobs Schedule completed jobs status every 2hours.
2	PSP JSS View 1. Successful Jobs 2. Failed Jobs	PSP JSS View	
3	PSP Service Metrics 1. Response Time 2. No of Requests	PSP Service Metrics	
4	PSP Adapters 1. Response Codes	PSP Adapters	
5	PSP API Gateway 1. TP Numbers (Check - Not Populating) 2. Response Codes	PSP API Gateway	
6	PSP Unknown Exceptions	PSP Unknown Exceptions	
7	Workforce	Workforce	

PAPI

S. No	Dashboard	Link
1	PAPI Gateway 1. TP Numbers (Check - Not Populating) 2. Response Codes	PAPI Gateway
2	PAPI GraphQL Gateway	PAPI GraphQL Gateway

Pubsub

S. No	Dashboard	Link
1	Pubsub (TODO : To enhance to capture all entities)	Pubsub Entity Publish
2	Pubsub TID Analyzer	Pubsub TID Analyzer

Report Job Framework

	Dashboard	Link	Comments
1	RJF JSS View 1. Successful Jobs 2. Failed Jobs	RJF JSS View Splunk Dashboard : QBDT PSP Report Jobs IKS	Update RJF Jobs Schedule completed jobs status every 2hours.

2	Jobs during the tiptoe window	100k_Query Assisted_Activation_Report Assisted_Agency_aid_rpt_SIT Assisted_Agency_aid_rpt_SUI Assisted_Agency_Info_Rpt_Others Assisted_Agency_Info_Rpt_SIT Assisted_Agency_Info_Rpt_SUI ASSISTED_DISCO_CUSTOMER_LIST_NEW Assisted_On_Hold_Info Assisted_Usage_Billing Back_Dated_Paychecks Back_Dated_voids Bad_EE_Accounts Bad_Funding_Account Bank_Report Cancelled_By_MissedPayrollsBatchJob Complia_Adjustments dd_limit_inf_jirapsp_6681 Duplicate_Adjustments EU_activ_err_count_exceed_limit Eu_Err_Aging_Info Financial_Ledger Financial_Transaction Foreign_Routing_Number Late_Payment_Details Late_Payment_Stats Manual_DataSync_Rep Manual_Ledger_Entry_Report Missed_Payments_High_Priority Missed_payments_not_immediate mmt_rec_count_exceed ms_m89_payment_info New_Signups_Report Psp_Cancelled_Assisted_Info PSP_COMPANY PSP_contact PSP_Entitlement_AppUser PSP_Entitlement_DBUser PSP_SOX_AppUser PSP_STATEID	
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PayrollDM Monitoring

S. No	Dashboard	Link
1	QBDT PSP DG2.0	QBDT PSP DG2.0

Performance

Traffic	Category	Dashboards
OLTP	Workflow Performance	OLTP Workflow Performance Comparison
	Query Performance	OLTP Query Performance Comparison
OLAP	Batch Job Performance	Batch Jobs Performance Comparison
	Query Performance	Batch Jobs Query Performance Comparison

Hourly Monitoring

Dashboards
PSP Overall Metrics - Hourly
PSP API Gateway Metrics - Hourly
PSP CPU and Memory Usage

Tiptoe Batch Jobs Monitoring

S No	Jobs	Validation Scripts
1	AMO	
2	Entity Event	
3	EmailGateway	
4	EMSBSToBRMDataSyncProcessor	
5	IAMEmailAddressProcessor	
6	CheckPrint	
7	EntitlementProcessor	
8	QbdtUnprocessedRequestsRetry	
9	WorkersCompProcessor	
10	AssistedUsageDataSyncProcessor	
11	Ledger Balance	

BatchJobs in Activity Window

S. No	Jobs	Comments
1.	AccountServiceSyncExceptionProcessor AMOMessageProcessor AssistedUsageDataSyncProcessor ATFDDataExtractProcessor BRMUsageErrorFileProcessor BulkWorkforceInviteProcessor CheckPrintProcessor EmailGatewayProcessor EntitlementProcessor EntityEventProcessor EntityEventRetryProcessor EntityInitialLoadProcessor EVSCompanyProcessor GemsGeneralLedgerProcessor IamEmailAddressProcessor LedgerBalanceProcessor LedgerOperationsProcessor MonthlyFeeProcessor MtlTransactionReportEnrichProcessor PISqlJobsProcessor Processor PSPTToEMSBSDDataSyncProcessor RealTimeEntityEventRetryProcessor RetryQbdtUnprocessedRequestProcessor RTBAutomationProcessor ScheduledEmailsProcessor SendCustomEmailsProcessor SendMonthlyDataToTFSPProcessor SoxDBUsersProcessor SoxReportProcessor StateReportProcessor WorkersCompProcessor	

Critical Batch Jobs Monitoring

S. No	Steps	Jobs
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1	Monitor Critical Batch Jobs.	- EFTPS All - Ledger Balance - EmailGateway - EMSBSToBRMDataSyncProcessor - CheckPrint - EntitlementProcessor - QbdtUnprocessedRequestsRetry - WorkersCompProcessor - AssistedUsageDataSyncProcessor
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Oracle Hash Specific Monitoring

S.No	Dashboard	Comments
1	DT Payroll DB Connection Info	
2		
3		

Workflows Monitoring

S. No	Workflow/AppName	Validation Points
1	Run Payroll	Hoolimath, Praveenkumar will run from test company.
2	SAP Monitoring	Singhal, Vikas to update script here
3	QBDT/QBDTWS Monitoring	Swain, Amrit to update script here

DB Queries Monitoring

S. No	Dashboard	Link
1	PSP DB Telemetry	Monitor long running queries. PSP DB Telemetry

Database Monitoring

S.No.	Dashboard	Link
1	Monitor Oracle Range --> Oracle Hash Replication Lag	GoldenGate Replication Dashboard
2	Ensure Veridata validation is green from Oracle Hash to Oracle Range	veridata console
3	Monitor Connections and Long Running Queries	
4	Oracle Hash Specific Dashboards	
5	Oracle Hash database dashboard	Oracle Hash Database Performance Monitor

Infrastructure Monitoring

Golden Gate Signals

Application	Module	Golden Gate Signals Link	Owner
PSP	QBDT Adapter	QBDT Adapter Golden Signals	Bansal, Jatin
	ADE Adapter	ADE Adapter Golden Signals	Bansal, Jatin
	BRM Adapter	BRM Adapter Golden Signals	Bansal, Jatin

	DIS Adapter	DIS Adapter Golden Signals	Bansal, Jatin
	EWS Adapter	EWS Adapter Golden Signals	Bansal, Jatin
	PTC Adapter	PTC Adapter Golden Signals	Bansal, Jatin
	CDM Adapter	CDM Adapter Golden Signals	Bansal, Jatin
	Batch Jobs	Batch Jobs Golden Signals	Bansal, Jatin
PAPI	PAPI	PAPI Golden Signals	Rajoria, Himanshu
RJF	RJF	RJF Golden Signals	Joshi, Vishwanath

Wavefront

Application	Wavefront Monitor	Owner
PSP	PSP Wavefront Monitor	Bansal, Jatin
PAPI	PAPI Wavefront Monitor	Rajoria, Himanshu
Pubsub	Pubsub Wavefront Monitor	Agarwal, Akshit
RJF	RJF Wavefront Monitor	Joshi, Vishwanath
PayrollIDM	PayrollIDM Wavefront Monitor	Agarwal, Akshit

AppDynamics

This is the backup for Infrastructure monitoring.

Environment	App Dynamics Link	Owner
Production	DTPayroll AppDynamics BaseURL	Agarwal, Akshit

Failed/Missed Batch Jobs

Validation queries for the jobs in the tiptoe window can be found [here](#)

App	Category	Job Names / Screenshots (Tiptoe)	Job Names / Screenshots (Rollback)
Batch Jobs	Scheduled jobs	EntitlementProcessor LedgerOperationsProcessor AMOMessageProcessor PSPToEMSBSDDataSyncProcessor EntityEventProcessor RetryQbdtUnprocessedRequestProcessor EmailGatewayProcessor AssistedUsageDataSyncProcessor CheckPrintProcessor PISqlJobsProcessor WorkersCompProcessor BulkWorkforceInviteProcessor RealTimeEntityEventRetryProcessor EVSCompanyProcessor RTBAutomationProcessor IamEmailAddressProcessor EntityInitialLoadProcessor MtlTransactionReportEnrichProcessor EntityEventRetryProcessor SoxDBUsersProcessor StateReportProcessor AccountServiceSyncExceptionProcessor SendCustomEmailsProcessor ScheduledEmailsProcessor LedgerBalanceProcessor	
	Missed Jobs		
	Failed Jobs		

RJF	Scheduled jobs	100k_Query Assisted_Activation_Report Assisted_Agency_aid_rpt_SIT Assisted_Agency_aid_rpt_SUI Assisted_Agency_Info_Rpt_Others Assisted_Agency_Info_Rpt_SIT Assisted_Agency_Info_Rpt_SUI ASSISTED_DISCO_CUSTOMER_LIST_NEW Assisted_On_Hold_Info Assisted_Usage_Billing Back_Dated_Paychecks Back_Dated_voids Bad_EE_Accounts Bad_Funding_Account Bank_Report Cancelled_By_MissedPayrollsBatchJob Complia_Adjustments dd_limit_inf_jirapsp_6681 Duplicate_Adjustments EU_activ_err_count_exceed_limit Eu_Err_Aging_Info Financial_Ledger Foreign_Routing_Number Manual_DataSync_Rep Missed_Payments_High_Priority Missed_payments_not_immediate mmt_rec_count_exceed ms_m89_payment_info New_Signups_Report Psp_Cancelled_Assisted_Info PSP_COMPANY PSP_contact PSP_Entitlement_AppUser PSP_Entitlement_DBUser PSP_SOX_AppUser PSP_STATEID	
	Missed Jobs		
	Failed Jobs		